whenever $||\underline{x} - \underline{a}|| < \delta$, then $|||f(\underline{x}) - \underline{l}|| < \varepsilon$

(Note: both $f(\underline{x})$ and $\underline{l}$ are members of R^m, not members of R^n)

Continuity of a function at a point of R^n

A $f: D \rightarrow R^m$, where $D \subseteq R^n$ and the point $\underline{a} = (a_1, a_2, \ldots, a_n) \in R^n$, we say f is continuous at $\underline{x} = \underline{a}$ if

i) $\lim_{\underline{x} \to \underline{a}} f(\underline{x})$ exits, *and* (ii) $\lim_{\underline{x} \to \underline{a}} f(\underline{x}) = f(\underline{a})$.

D) **Extreme Value Theorem**

The statement and discussion of the theorem requires introduction of a number of concepts, first for the functions of *single variable*, and then for functions of *several variables*.

We have already discussed in Block 3, and will discuss in detail in Block 5 and 7 how the concept of derivative (from differential calculus) is useful in solving optimization problems, i.e. problems concerned with minimization of, say, cost of production depending upon many variables representing inputs ingredients (including human labour); and maximization of profits. In this context, we recall some of the relevant concepts about functions of single variable and then extend these concepts to functions of several variables.

-Local minima, local maxima, and local extrema (extreme value) and Extreme Value Theorem for *single* independent variable

The above-mentioned concepts are discussed for a *function (or dependent variable) z =f(x), dependent on independent variables say x*, y etc. *Here, first we discuss these concepts for a single independent variable, say, x over a set S of real numbers.* Then we extend the concepts to more than one variable.

In analysis/calculus, the concept of '*local' for a real number p* is its immediate neighbourhood, which, without loss of generality, may be taken as the open interval $]\,p-\varepsilon, p+\varepsilon\,[$, for any $\varepsilon > 0$.

Local Minima: a local minimum value, also called a *local minima,* of a function (or dependent variable) z =f(x), defined on subset S of R, *is the value that z attains* at some point p such that for $\varepsilon > 0$, *and* $\forall\, x \in\,]\,p-\varepsilon, p+\varepsilon\,[\subseteq S$, we have $f(x) > f(p)$. *This is true for the value corresponding to value* $x = x_3$ in Fig 13.2.

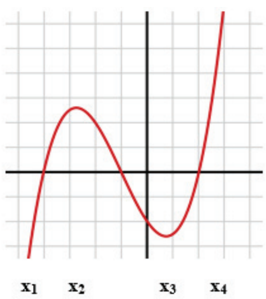

Fig. 13.2:

It may be noted that $x = x_3$ is a *point of local minima*, but it is NOT a local minima. *A local minima is the value $f(x_3)$.* There may be more than one local minima as shown in **Fig. 13.3.**

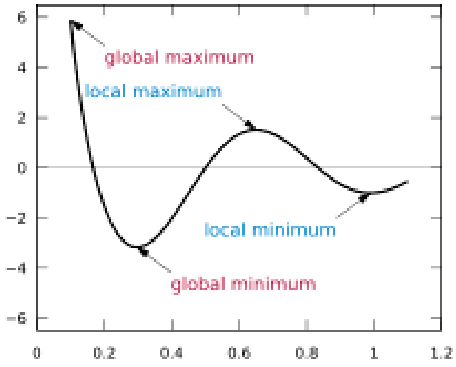

Fig. 13.3:

Local Maxima: *a* local maximum value, also called *a local maxima,* of a function (or dependent variable) z =f(x), defined on subset S of R, *is the value that z attains* at some point p such that for $\varepsilon > 0$, *and* $\forall$ x $\in$] $p-\varepsilon$, $p + \varepsilon$ [$\subseteq$ S, we have *$f(x) < f(p)$. This is true for the value corresponding to $x = x_2$*in the Fig.13.2

It may be noted that $x = x_2$ is a ***point of local maxima***, but it is NOT a local maxima. ***A local maxima is the value $f(x_2)$.*** There may be more than one local maxima as shown in figure 13.3.

Local Extremum/Extrema: A local minimum or a local maximum is called a *Local Extremum (Plural: extrema)*

Global/ Absolute Minimum: A value m of a function f(x) over a set S is said to be *the* Global/ Absolute Minimum *(generally, the term 'minima' is not used here.)*, if $m \leq f(x)$, $\forall$ x $\in$ S, e. g. the value corresponding to $x = x_1$ in Fig. 13.2. In Fig.13.3, it is shown in bottom red.

Global/ Absolute Maximum: A value M of a function f(x) over a set S is said to be ***the*** Global/ Absolute Maximum (generally, the term 'maxima' is not used here.), if $f(x) \leq M$,$\forall$ x $\in$ S, e. g. the value corresponding to $x = x_4$ in Fig. 13.2. In Fig. 13.3, it is shown in top red.

Remarks: It may be noted that the global maximum (or global minimum) may not be even a local maxima(or local minimum), i.e., global maximum (global minimum) may not be largest value (least value) *for values at neighbouring points*. For example, in Fig 13.2, for points at x_4+ (point right to x_4), the value may exceed the value at x_4. Similarly, in Fig. 13.2, for points at x_1– (point left to x_1), the value may be less than the value at x_1.

In general, the global maximum (global minimum), (only) at the boundary points, may not be a local maximum (local minimum).

Extreme Value Theorem for single variable states that for a function *f* of one variable, if *f* is continuous on a closed interval [*a*, *b*], then *f* has an absolute maximum value and an absolute minimum value in [a, b].

- Local minima, local maxima, and local extrema (extreme value) for *multiple/several* independent variables.

In the previous subsection, the above-mentioned concepts were discussed *for a single independent variable, say, x over a set S of real numbers***.** Here we extend these concepts to the case of *several variables*.

In analysis/ calculus, the concept of '*local' for* $p \in R^n$

is its immediate neighbourhood, which, without loss of generality, may be taken as the *open disc*of radius $\varepsilon > 0$.

Local Minima: *a* local minimum value, also called *a local minima,* of a function (or dependent variable) z =f($\underline{x}$), defined on subset S of R^n, for $\underline{x} = (x_1, x_2, \ldots, x_n)$ $\in$ *S, is the value that z attains* at some point $\underline{p} \in R^n$, such that for $\varepsilon > 0$, *and* $\forall$ x $\in$ open disc centered at p of radius $\varepsilon \subseteq S$,we have $f(x) > f(p)$.

It may be noted that *the point* $\underline{p}$ is NOT a local minima. *A local minima is the value f(*$\underline{p}$*).* There may be more than one local minima.

Local Maxima: *a* local maximum value, also called *a local maxima,* of a function (or dependent variable) z =f($\underline{x}$), defined on subset S of R^n, for $\underline{x} = (x_1, x_2, \ldots, x_n)$ $\in$ *S, is the value that z attains* at some point $\underline{p} \in R^n$, such that for $\varepsilon > 0$, *and* $\forall$ x $\in$ open disc centered at p of radius $\varepsilon \subseteq S$, we have $f(x) < f(p)$.

It may be noted that *the point* $\underline{p}$ is NOT a local maxima. *A local maxima is the value f(*$\underline{p}$*).* There may be more than one local maxima.

Local Extremum/ Extrema: A local minimum or a local maximum is called a *Local Extremum (Plural: extrema)*

Global/ Absolute Minimum: A value m of a function f($\underline{x}$) over a set $S \subseteq R^n$ is said to be the Global/ Absolute Minimum *(generally, the term 'minima' is not used here.)*, if $m \leq f(\underline{x})$, $\forall \underline{x} \in S \subseteq R^n$.

Global/ Absolute Maximum: A value M of a function f($\underline{x}$) over a set $S \subseteq R^n$ is said to be the Global/ Absolute Minimum *(generally, the term 'maxima' is not used here.)*, if $f(\underline{x}) \leq M$, $\forall \underline{x} \in S \subseteq R^n$.

In Unit 15, we will discuss how to calculate maxima and minima of functions of several variables.

Extreme Value Theorem for several variable states that for a function *f* of one variable, if *f* is continuous on a closed disc D, then *f* has an absolute maximum value and an absolute minimum value in D.

E) **Intermediate Value Theorem**

In mathematical analysis, the intermediate value theorem states that if is a continuous function whose domain contains the interval [a, b], then it takes on any given value between and at some point within the interval.

13.6 LET US SUM UP

In this unit we learnt about the concepts of sequence of Real numbers, and discussed its special types viz. bounded sequences, monotone sequences, followed by the concepts of Limit of Sequence, Convergence, Divergence and Cauchy sequences, Sub-sequences, Sequence in R^n. For the subsets of real numbers, we introduced the concepts of Open Sets, Closed Sets, Bounded Sets, Compacts Sets, Bolzano-Weierstrass Theorem and discussed their Economic Applications. Finally, under Analysis of Several Real Variables, we discussed the n-dimensional real space, n-dimensional Euclidean Space, Functions between Euclidean Spaces, Function from $R^n \; to \; R$, Functions from R^k to R^m.

13.7 KEY WORDS

Real analysis : Real analysis is the branch of Mathematical Analysis

Sequence : A Sequence may be considered as an enumerated ordered set, in which repetition of elements is allowed. Enumerated means elements can be counted as first element, second element and so on. Elements of the interval [0, 1] cannot be counted as first, second, and so on. A sequence is bounded by pair of parentheses, i. e. '('and ')', unlike a set which the boundaries of which are braces, e. g. '{'and '}'.

Sequence of real numbers : A *sequence* of real numbers is a function f: S → R, where S, the ***domain,*** is either the set N of natural numbers, or an interval subset of N, and R, the ***codomain***, is the set of real numbers. In the definition of a sequence as a function, our main interest is in the range of the function f, as an enumerated ordered set allowing repetitions.

Convergent Sequence : A sequence (S_n) is said to be convergent *if* ∃a (finite) real number l, such that for given $\varepsilon > 0$, ∃*a natural number* n_ε such that $|S_n - l| < \varepsilon$, *for all* $n > n_\varepsilon$. *Then l is called the limit of the sequence (S_n), and we write* $\lim_{n\to\infty} Sn = l$.

Cauchy Sequence : A sequence (S_n) is said to be a Cauchy's sequence *if* for given $\varepsilon > 0$, ∃ *a natural number* n_ε such that $|S_n - S_m| < \varepsilon$, *for all* $n, m > n_\varepsilon$.

Limit-point of a sequence : For a sequence (S_n), a real number, say, *LP* is said to be a limit point of the sequence (S_n), if some *subsequence* of it converges to *LP*.

Open set : A set S of real numbers is said to be *open*, if for each element *a* of S, there is a number ε >0 such that the open interval]a– ε, a+ ε[⊆S. In other words, every member of S is *well inside* S, and hence, no member is a boundary point.

Closed set : A set S ⊆ R is said to be a closed if R – S = (–∞, + ∞) – S (the set of real numbers not in S) is open. R, the set of real numbers is also denoted as open interval (–∞, + ∞).

Compact set : A subset of R is called ***compact***, if it is both (i) closed, and (ii) bounded.

Limit point of a set : A real number p is called a limit point of *set* S of real numbers if for every *ε >0* the open interval] p–*ε, p + ε* [contains a point of S different from p. The Limit point of a set is also called accumulation/ condensation/ cluster point of a set.

13.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) a) It is subsequence, as each of its terms is in S in the same order.

b) It is NOT a subsequence, as its terms 12 and 6 are not in the same order as in S.

c) It is NOT a subsequence, as its term 243 is not in S.

2) a) (0, 3/2, 2/3, 5/4, 4/5

b) $((1/2)^{1/2}, (2/3)^{1/3}, (3/4)^{1/4}, (4/5)^{1/5}, (5/6)^{1/6})$

3) a) 11/ (6n – 4)

b) $2^n - 3$.

4) a) $S_n = 3 S_{n-1} + 1$, and $S_1 = 1$ is the required recursive definition.

b) $S_1 = 4$, and $S_n = (n+1)^2$, $S_{n-1} = n^2$. Therefore, $S_n - S_{n-1} = (n+1)^2 - n^2 = 2n + 1$

Hence, $S_n = S_{n-1} + 2n + 1$is the required recursive definition.

Check Your Progress 2

1) a) The A.P has the initial term 7 and common difference is – 2.5. Therefore,

$t_{25} = 7 + (25-1) \times (-2.5) = 7 - 60 = -53.$

b) The required sum = $t_{25} - t_{14} = -53 - [7 + (14-1) \times (-2.5)] = -53 - (-25.5) = -27.5$

2) a) The given sequence is a G.P with initial term a= 6, and common ratio as (*1/3*). Substituting these values in*a* $(r^n - 1)/(r-1)$, we get

$S_{10} = 6 (1-(1/3)^{10})/(1-(1/3)) = 9 ((1-(1/3)^{10})$

b) i) The G.P has initial term $a = 18$, and the common ratio $r = (-1/3)$.

Therefore,

$t_{15} = 18 (-1/3)^{(15-1)} = 18(1/3)^{14} = 2 (1/3)^{12} = \ldots$

ii) The required sum = $S_{15} - S_8 = [18/(1-((-1/3))] \times [\{(1-(-1/3)^{15})\} - \{(1-(-1/3)^{8})] =$

$= [(18 \times 3)/4] \times [(1/3)^{15} + (1/3)^{8}] = (27/2) [(1/3)^{15}(1 + 3^7)$

$= (1/2) (1/3)^{12} (1 + 3^7).$

3) Thenth term of A.P is (5– 2n)/ 30, n = 1, 2, 3, …

Therefore, we get the corresponding H. P. as *30/ (5– 2n)*, n = 1, 2, 3, …,

or, as 10, 30, – 30, – 10, – 6, …

4) a) First 12 terms are, 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89.

b) The required sum = (sum of first10 terms) – (sum of first 5 terms)

$= (F_{12} - 1) - (F_7 - 1) = F_{12} - F_7 = 89 - 8 = 81.$

5) a) $(1, 2, 4, 8, \ldots, 2^{n-1}, \ldots) = (2^{n-1})$ for $n \in N$

b) $(-1, -2, -4, -8, \ldots, -2^{n-1}, \ldots) = (-2^{n-1})$ for $n \in N$

c) (2, 3, 2, 3, ….); $t_n = 2$, if n is odd, and $t_n = 3$, if n is even

6) a) (2, 3, 2, 3, …); $t_n = 2$, if n is odd, and $t_n = 3$, if n is even

b) $(1, 2, 4, 8, \ldots, 2^{n-1}, \ldots) = (2^{n-1})$ for $n \in N$

c) $(3 - (1/n))$ for $n \in N$ is both bounded and monotonic.

Check Your Progress 3

1) a) (4, 3, 2, 1, 4, 3, 2, 1, …)

b) $(1, 4, 9, 16, \ldots) = (n^2)_{n \in N}$

c) $(-1, -4, -9, -16, \ldots) = (-n^2)_{n \in N}$

d) (1, – 1, 4, – 4, 9, – 9, 16, – 16, …)

2) a) It is apparent that terms t_n of the given sequence may be considered as constituted of the following four sub-sequences

i) $t_n = 4$ for all $n = 4m - 3$, for $m \in N$

ii) $t_n = 3$ for all $n = 4m - 2$, for $m \in N$

iii) $t_n = 2$ for all $n = 4m - 1$, for $m \in N$

iv) $t_n = 1$for all $n = 4m$, for $m \in N$

Each of the sub-sequence a., b., c., d. is a constant sequence, and converge respectively to 4, 3, 2, 1.

Hence the sequence has four limit points, viz. 4, 3, 2, 1.

b) The sequence (2+1, 3– 1, 2+1/2, 3– ½, 2+1/3, 3– 1/3, 2+1/4, 3– ¼, …) is constituted of the two sub-sequences

$(2+1, 2+1/2, 2+1/3, 2+1/4, \ldots) = (2+ (1/n)), n \in N$,

AND

$(3- 1, 3- ½, 3- 1/3, 3- ¼, \ldots) = (3 - (1/n)), n \in N$

Sub-sequence 'a.' converges to 2 as follows

For given $\varepsilon > 0$, howsoever small and $1 > 0$, by Archimedes Principle, there exists a natural number M

Such that $M. \varepsilon > 1$ implies $\varepsilon > 1/M > 1/n$ for all $n > M$.

Thus, $|(2+ (1/n)) - 2| = 1/n < 1/M < \varepsilon$ for all $n > M$.

Exactly on similar lines, we can show that subsequence 'b.' converges to 3.

3) $(n)^{(1/n)} > 1^{(1/n)}$, for all $n > 1$

Thus, $(n)^{(1/n)} = 1 + h_n$ *for some* $h_n \geq 0$. Or

$n = (1 + h_n)^n$, expanding by binomial theorem, we get

$n = 1 + n\, h_n + n\,(n-1)/2\,(h_n)^2 + \ldots$

(All the terms on R.H.S. are positive, neglecting some will reduce the number on R.H. S., we get)

$n > [n\,(n-1)/2]\,(h_n)^2$ i.e. $(h_n)^2 < 2/(n-1)$ for all $n \geq 2$, i. e. $|h_n| < \sqrt{(2/(n-1))}$ for all ≥ 2.

For given $\varepsilon > 0$, to find n_ε so that equation (ii) below is satisfied by all $n > n_\varepsilon$

$|h_n| < \sqrt{(2/(n-1))} < \varepsilon$, … (ii)

ii) is true if $\mathbf{2} / \varepsilon^2 < (\mathbf{n}-1)$ *or* $(\mathbf{2} / \varepsilon^2) + 1 < n$

Taking $n_\varepsilon = (\mathbf{2} / \varepsilon^2) + 1$, *we*

$|n^{(1/n)} - 1| < |h_n| < \varepsilon$ *for all* $n > (2 / \varepsilon^2) + 1$.

4) We prove the result by contradiction. Let $l > 0$ be the limit of (n).

Then for given $\varepsilon = 1/2$, there must exist n_ε such that

$|\mathbf{n} - l| < \varepsilon = 1/2$ for all $n > n_\varepsilon$, i.e.

For the limit to exist, we must have

$l - 1/2 = l - \varepsilon < n < l + \varepsilon = l + 1/2$ for all $n > n_\varepsilon$

However, there can be at most one natural number in the interval $]l - ½, l + 1/2[$.

Therefore, there cannot be any n_ε satisfying the requirement for existence of the limit l.

Check Your Progress 4

1) a) Let [a, b] be a closed set. Then for any $\varepsilon > 0$, *the interval* $]a - \varepsilon, a + \varepsilon[\not\subseteq [a, b]$. Hence, [a, b] is not an open set. Similarly,

b) If the semi-closed interval is [a, b[then as above, for any $\varepsilon > 0$, *the interval* $]a - \varepsilon, a + \varepsilon[\not\subseteq [a, b[$. If the semi-closed set is]a, b], then as above, for any $\varepsilon > 0$, *interval* $]b - \varepsilon, b + \varepsilon[\not\subseteq]a, b]$.

c) (i) is not bounded below, but is bounded above with one of upper bounds being 10,

(ii) is not bounded above, but is bounded below, with one of the lower bounds being 100.

d) We have already shown every finite set is closed (Refer 13.4.1 Example (ii)), and bounded (refer the exercise section, exercise 6)

2) a) The set N= {1, 2, 3, …} has no limit points because it cannot contain any interval$(a - \varepsilon, a + \varepsilon)$ of real numbers.

b) The set of limit points of $(3, 5] \cup \{6, 7\} \cup [8, 9]$ is $[3, 5] \cup [8, 9]$, because every point of an interval is its limit point. Also, a boundary point is its limit point. Further, a finite set has no limit point.

Check Your Progress 5

1) As the function is defined at each the values of x, y and z, therefore, the *domain* is

$R^3 = R \times R \times R$.

Also, as every real value can be assumed (because for any u of R the value of f (u/(– 3), 0, 1) =u), hence, the range of f =R.

a) $f(-1, -2, 2) = 2(-1)(-2)^2 \times 2 + 2(-2)^3(2)^2 - 3(-1)2 = -16 - 64 + 6 = -74$.

b) $f(1, 3, -2) = 2(1)(3)^2(-2) + 2(3)^3(-2)^2 - 3(1)(-2) = -36 + 216 + 6 = 186$.

2) a) $||\underline{x}|| = ||(4, 2, -1, -2, -3)|| = \sqrt{[(4)^2 + (2)^2 + (-1)^2 + (-2)^2 + (-3)^2]}$

$= \sqrt{[16 + 4 + 1 + 4 + 9]} = \sqrt{34}$.

$||\underline{y}|| = ||(-1, 3, -3, -3, 2)|| = \sqrt{[(-1)^2 + (3)^2 + (-3)^2 + (-3)^2 + (2)^2]}$

$= \sqrt{[1 + 9 + 9 + 9 + 4]} = \sqrt{32}$

b) Euclidean distance between

$\underline{x} = (4, 2, -1, -2, -3)$, and $\underline{y} = (-1, 3, -3, -3, 2)$

is $d(\underline{x}, \underline{y}) = \sqrt{[(4-(-1))^2 + (2-3)^2 + (-1-(-3))^2 + (-2-(-3))^2 + ((-3)-2)^2]} = \sqrt{[25 + 1 + 4 + 1 + 25]} = \sqrt{56}$

c) scalar product $\underline{x}.\underline{y} = 4(-1) + 2(3) + (-1)(-3) + (-2)(-3) + (-3)(2)$

$= -4 + 6 + 3 + 6 - 6 = 5$

13.10 EXERCISES AND ANSWERS

Q 1. Show that an Arithmetic Progression is recursively definable.

Ans.

If a is the initial term and d, the common difference, then nth term is given by

$t_n = a + (n-1) \times d = a + [(n-2) \times d + d] = [a + (n-2)] \times d + d = t_{n-1} + d$, *i. e.*

$\boldsymbol{t_n = t_{n-1} + d}$; n^{th} term is obtained through a formula involving $(n-1)^{th}$ term.

Q2. *Show that* an A. P. $(a + (n-1)d) = (a, a+d, a+2d, \ldots)$, which is non-trivial (i. e., $d \neq 0$),

(i) is *not* bounded above, if $d>0$, and

(ii) is *not* bounded below if $d<0$.

Ans.

Proof: (i) Let $M>0$, and $M > a$, be a real number howsoever large. Then given real number $M-a$, *and* $d>0$, *by Archimedean Property,* $\exists$ (there exists) a natural number n such that $n\,d > M-a$, *or* $(a + nd) > M$. Hence $(n+1)^{th}$ term of the given A. P. will be greater than M, where M is as large as we like. Hence, A. P. cannot be bounded above.

Proof (ii) $d < 0 \Rightarrow -d > 0$. Let $M>0$, and $M > a$, be a real number howsoever large. Then given real number $M + a$, *and* $-d>0$, *by Archimedean Property,* $\exists$ a natural number n such that $n(-d) > M + a$, **or** $(-a - nd) > M$. Multiplying both sides by (-1) and reversing the inequality, we get

$(a + n\,d) < -M$. Thus $a + n\,d$ can be as large we like with *negative sign*. Hence, the A.P. cannot be bounded below.

Q3. *Show that* ***a*** G. P. $(a\,r^{(n-1)}) = (a, a\,r, a\,r^2, \ldots)$, which is non-trivial (i. e., $a\neq 0$, $1\neq r\neq 0$)

(i) It is ***not*** bounded above, if ($r>1$, and $a>0$),

(ii) It is ***not*** bounded below if $r>1$, *and* $a<0$)

iii) Bounded neither above, nor below if $r < (-1)$

Ans.

Proof (i) Let $r = 1+\varepsilon$, for $(a.\ \varepsilon) > 0$ and let $M > 0$ be a real number, as large as we like. By Archimedes property for $(a.\ \varepsilon) > 0$, $\exists$ natural number n such that $n.\ (a.\ \varepsilon) > M$. … (A)

$$\text{Also, } (a.\ (r)^n) = a.\ (1+\varepsilon)^n = a.\ (1 + n\,\varepsilon + \ldots) > (a.\ n\,\varepsilon) \ \ldots \text{(B)}$$

From (A) and (B), we get that $\exists$an n such that $(a.(r)^n) > M$. Hence, the G. P. $(a.\ r^n)$, with $r>1$ is not bounded above.

Proof (ii): As $a<0$, we have $(-a)>0$. *Also,* $r> 0$, let $= 1+\varepsilon$, *with* $\varepsilon > 0$, Then, for $[(-a).\ \varepsilon)] > 0$ and let $M > 0$ be a real number, as large as we like, then by Archimedes property, $\exists$ natural number n such that $n.\ ((-a).\ \varepsilon) > M$. … … … (A)

Also, $(-a.)\ (r)^n) = (-a.).\ (1+\varepsilon)^n = (-a.).\ (1 + n\,\varepsilon + \ldots) > ((-a.).\ n\,\varepsilon)$ … (B)

From (A) and (B), we get that $\exists$an n such that $((-a.).(r)^n) > M$. Multiplying both sides by (-1) and reversing the inequality, we get

$(a.(r)^n) < -M$, where M is as large as we like, but with negative sign.

Hence, the G.P $(a.\ r^n)$, with $r>1$, and $a<0$, is not bounded below.

Proof (iii): $r < (-1)$, *then* $r^2 > 1$.

Consider two sub-sequences $\boldsymbol{(a\ r^{2(n-1)})}$, and $\boldsymbol{(a\ r^{2n-1})}$. There is no loss of generality, if we assume $a>0$.

Then, the subsequence $(a\,r^{2(n-1)})$*, by putting* $R_1 = r^2 > 0$*, becomes the sequence* $(a\,R_1^{(n-1)})$*, and the sequence is unbounded above by argument of* **(i)**

The subsequence $\boldsymbol{(a\ r^{2n-1})}$*, by putting* $R_1 = r^2 > 0$*, and* $A_1 = a.\ r < 0$ *becomes the sequence* $(A_1\,R_1^{(n-1)})$*, and the sequence is unbounded below by argument of* **(ii).**

Q4. *Show that a* Fibonacci sequence $(F_n) = (0, 1, 1, 2, 3, 5, 8\ldots)$ is not bounded above, but is bounded below by 0.

Ans.

By definition

$F_0 = 0$, $F_1=1$, and $F_n = F_{n-1} + F_{n-2}$, for $n \geq 2$. … … (I)

By mathematical induction, it is easily seen that for all $n > 7$

$F_{n+1} > F_n > n$ … … (II)

As $F_8 = 13 > 8 = F_7 > 7$. Therefore, for n=7, (II) is true.